

Digilians

Data Analysis

Data Analytics

- **Data Analytics** is a process of examining, cleaning, transforming and interpreting data to discover useful information, draw conclusions and support decision-making.
- It helps businesses and organizations understand their data better, identify patterns, solve problems and improve overall performance.

What is Data Analytics

The process of analyzing raw data to make conclusions about that information.







Data Analytics Jobs Opportunities



Tools & Skills for Data Analytics

- To strong skill for Data Analysis we needs to learn this resources to have a best practice in this domains.
- Python For Data Analytics
- SQL For Data Analytics
- Excel for Data Analytics
- Power BI / Tableau
- Mathematics & Statistics for Data Analysis

Data Analysis Libraries

- Gain hands-on experience with the most powerful Python libraries:
- Pandas: Data manipulation and analysis
- NumPy: Numerical operations and matrix handling
- Matplotlib/Seaborn: Data visualization
- Scikit-learn: Data preprocessing and statistical modeling

Understanding the Data

- Before starting any analysis it's important to understand the type and structure of your data. This helps you choose the right methods for cleaning, exploring and analyzing it.
- What is Data?
- Sample vs Population
- Qualitative vs Quantitative Data
- Univariate vs Multivariate Data
- Nominal, Ordinal, Interval and Ratio Scales

Reading and Loading Datasets

- Reading and Loading Datasets is the first step in data analysis where you import data from files like CSV, Excel or databases into your working environment such as Python or Excel so you can explore, clean and analyze it.
- Reading CSV, Excel and JSON files
- Exporting dataframes to CSV/JSON
- Slicing, Indexing, Manipulating and Cleaning DataFrames

Data Preprocessing

- Data preprocessing involves cleaning and transforming raw data into a usable format. It includes handling missing values, removing duplicates, converting data types and making sure the data is in the right format for accurate results.
- Data Preprocessing
- What is Data Cleaning?
- Handling Missing Data
- Handling outliers
- Data Transformation
- Feature Engineering
- Data Sampling

Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) in data analytics is the initial step of analyzing data through statistical summaries and visualizations to understand its structure, find patterns and prepare it for further analysis or decision-making.
- Exploratory Data Analysis in Python

Univariate Analysis

- Measures of Central Tendency
- Measures of spread, IQR
- Skewness & Kurtosis
- Visualization: Histograms, Boxplots, Q-Q plots

Multivariate Analysis

- Correlation and Covariance
- Cross-tabulation
- Cluster Analysis, MANOVA(Multivariate Analysis of Variance), Factor and Canonical Correlation Analysis

Data Visualization

- Data visualization uses graphical representations such as charts and graphs to understand and interpret complex data.
- What is Data Visualization and Why is It Important?
- Visualization with Matplotlib
- Visualization using Seaborn
- Visualization using Plotly
- PowerBI and Tableau

Probability & Statistics in Data Analytics

- It help you understand data, find patterns and make smart decisions. Probability deals with chances and likelihoods, while statistics helps you collect, organize and interpret data to see what it tells you.
- Probability Distributions
- Central Limit Theorem
- PDF vs CDF
- Confidence Intervals
- Z-score, T-distribution
- P-Values & Hypothesis Testing
- One-Tailed vs Two-Tailed Tests
- Chi-Squared Tests
- Point Estimation

Time Series Data Analysis

- Time Series Data Analysis is the process of studying data points collected or recorded over timelike daily sales, monthly temperatures or yearly profits to find patterns, trends and seasonal changes that help in forecasting and decision-making.
- Define Time Series Data
- Data and Time function in Python
- Time Series Data Plotting
- Deal with missing values in a Time series
- Moving Averages : Stationarity, Seasonality, Trend
- Augmented Dickey-Fuller Test
- Autocorrelation

What is Data

- Data is the raw form of information, a collection of facts, figures, symbols or observations that represent details about events, objects or phenomena.
- By itself, data may appear meaningless, but when organized, processed and interpreted,
- it transforms into valuable insights that support decision-making, problem-solving and innovation.

Importance of Data

- **Decision-making and insights:** Organizations use data to make better decisions. Raw data becomes useful when transformed into insights with the help of analytics.
- **AI/ML and Innovation:** Data is the fuel for artificial intelligence and machine learning. More and higher-quality data means better training, more accurate predictions.
- **Digital transformation:** The rise of Big Data has enabled new capabilities i.e from real-time analytics to personalized services.

Types of Data

- Data can be categorized in different ways depending on how it is collected, stored and represented. Broadly, it falls into the following types:

Types of Data

- 01 Quantitative Data
- 02 Qualitative Data
- 03 Structured Data
- 04 Unstructured Data
- 05 Semi-Structured Data

1. Quantitative Data

- Quantitative data is information that can be measured, counted and expressed in numerical form.
- It provides objective values that can be analyzed statistically to identify patterns, trends and relationships.

- Represents numbers and measurable values.
- Can be divided into: Discrete data (Whole numbers) and Continuous data (Values on a scale).
- Widely used in research, finance, engineering and business analytics.
- **Example:** Age of people, number of customers visiting a store, temperature readings, sales revenue.

2. Qualitative Data

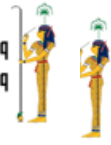
- Qualitative data is descriptive, non-numeric information that explains qualities, characteristics or categories rather than quantities.
- It helps understand opinions, experiences and meanings behind behaviors.

- Focuses on qualities, attributes and categories rather than numbers.
- Often collected through surveys, interviews or observations.
- Useful for understanding opinions, motivations and behaviors.
- **Example:** Customer feedback (“satisfied”, “unsatisfied”), product colors, interview transcripts, social media comments.

3. Structured Data

- Structured data is information organized into a predefined format (rows and columns) that makes it easily searchable and manageable by traditional databases.

- Stored in relational databases or spreadsheets.
- Easy to process with SQL and other tools.
- Best suited for tasks requiring accuracy and consistency.
- **Example:** Bank transactions, employee records, product inventories.



4. Unstructured Data

- Unstructured data is raw information that does not follow a predefined structure or format making it harder to organize and analyze with conventional tools.

- Accounts for over 80% of data generated globally.
- Requires advanced tools (AI, NLP, computer vision) to extract insights.
- Common in social media, multimedia and IoT applications.
- **Example:** Emails, images, videos, voice recordings, PDF documents.

5. Semi-Structured Data

- Semi-structured data combines aspects of structured and unstructured data.
- It does not reside in traditional tables but still contains tags or markers that provide a loose structure.

- Provides a balance between flexibility and structure.
- Easier to analyze than unstructured data, but less rigid than structured data.
- Often used in web applications, IoT devices and log systems.
- **Example:** JSON files, XML documents, NoSQL databases, sensor logs.

✓ 1. Quantitative Data – Sample Dataset

Student	Age	Height (cm)	Score
A	18	165	82
B	20	170	90
C	19	160	75
D	22	175	88

✓ 2. Qualitative Data – Sample Dataset

Person	Eye Color	Country	Car Brand
A	Brown	Egypt	Toyota
B	Blue	USA	BMW
C	Black	India	Kia
D	Green	France	Honda

✓ 3. Structured Data – Sample (Table from a Database)

Employee ID	Name	Position	Salary
101	Omar	Manager	20,000
102	Sara	Analyst	12,000
103	Hani	Developer	15,000
104	Lina	HR	10,000

✓ 4. Unstructured Data – Sample

These are raw, free-text or multimedia examples:

Sample 1 – Customer Review

"The product arrived late and the quality is not good. I will not buy again."

Sample 2 – Email

Subject: Meeting

"Hello team, we will meet tomorrow at 9 AM to discuss the project."

Sample 3 – Social Media Post

"I love this restaurant! Great food 🍷🔥"

Sample 4 – File types

- A selfie image
- A 2-minute video
- A voice note
- A PDF report

✓ 5. Semi-Structured Data – Sample

JSON Example

```
json

{
  "name": "Ahmed",
  "age": 25,
  "skills": ["Python", "SQL"],
  "city": "Cairo"
}
```

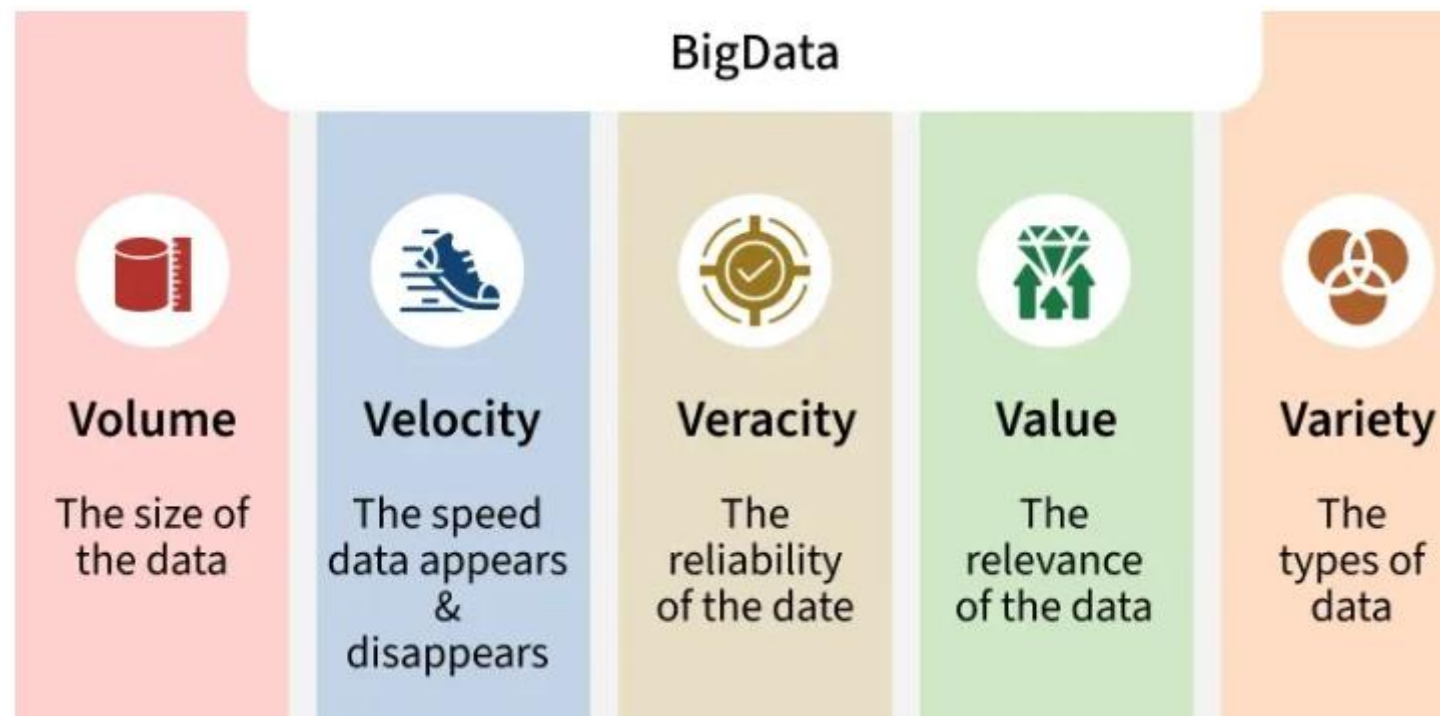
XML Example

```
xml

<student>
  <name>Mary</name>
  <age>21</age>
  <major>Computer Science</major>
</student>
```

Big Data

- When datasets grow in size, complexity and speed, traditional methods don't suffice.
- Big Data refers to datasets that are too large, too varied or too fast to be handled by traditional data processing tools.



The defining characteristics often called the Vs of Big Data are:

- **Volume:** Massive amounts of data.
- **Velocity:** Speed of generation and processing.
- **Variety:** Different formats: structured, unstructured, semi-structured.
- **Veracity:** Accuracy, trustworthiness of data to deal with noise and errors.
- **Value:** The usefulness of data i.e having data is not enough, we must extract value from data.

Data Collection

- Data collection is the process of acquiring data from various sources and in diverse formats for the purpose of storage,
- analysis and insight generation.
- It's often the first step in the data life cycle

- It helps ensure data is accurate and useful.
- Can be done manually or automatically using tools and software.
- Proper collection is the first step in making data meaningful.

Examples

- Collecting customer surveys, website clicks, sensor readings, social media posts.
- Bringing data from multiple sources into one place (databases, files, cloud).

Data Management

- Data management refers to all the practices, policies and technology used to collect, store organize, process, maintain and make data available in a secure, efficient and usable form.
- It covers the full lifecycle from creation to disposal

- Includes storing organizing and updating data.
- Ensures data quality, consistency and availability.
- Helps different teams access the right data for their work.
- **Examples**
- Storing company records in databases.
- Cleaning and organizing data to remove errors.

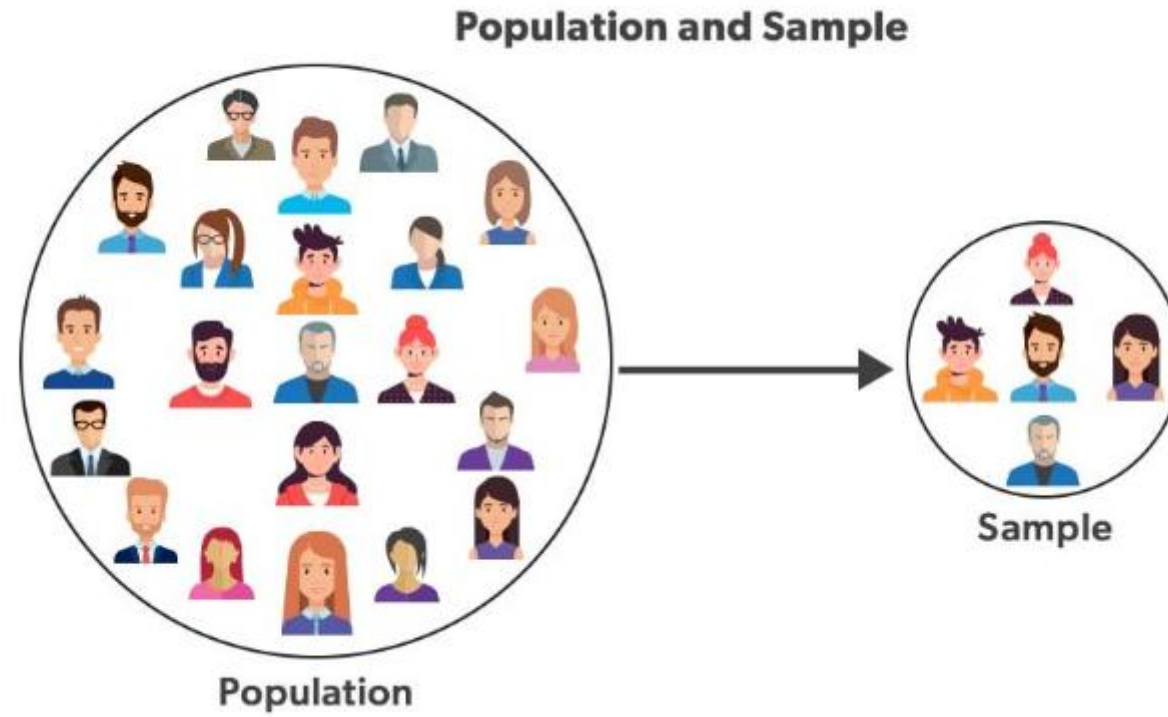
Data Security

- Data security refers to protecting data against unauthorized access, corruption, theft, loss or misuse.
- It involves both technical controls and policy or governance measures.

- Protects privacy and prevents misuse of data.
- Builds trust with customers and stakeholders.
- Prevents financial loss and reputational damage.
- **Examples**
- Using passwords and encryption to protect sensitive information.
- Limiting who can access customer records or financial data.

Population vs Sample in Statistics

- In statistics, **population** and **sample** are fundamental concepts used to describe groups of data:
- A population refers to the entire set of individuals, objects, or data points that you want to study. It can be large or small depending on the scope of your research.
- **For example**, all students in a school or all people in a country.
- It provides a complete picture and is usually denoted by **N**.



- A sample is a subset of the population that is selected for analysis.
- It's used when studying the entire population is impractical or impossible.
- Sampling allows for inferences about the population using statistical techniques.

- For example, if the population is all students in a school, a sample could be 50 students randomly chosen from different classes to participate in a survey.
- It offers an estimate and is denoted by n .
- Parameters (like population mean) describe the population, while statistics (like sample mean) describe the sample.
- Sampling enables us to make inferences about the population using statistical techniques.

Collecting Data From the Population and Sample

- **When to use a Population:**
- Populations are used when your research question requires it, or when you have access to data from every member of the population.
- Usually, it is only straightforward to collect data from a whole population when it is small, accessible, and cooperative.

Example:

- *A marketing manager at a **small local bakery** wants to understand customer preferences.*
- *They collect data on **every customer's bread purchase** over a month.*
- *Since the customer base is limited and accessible, they analyze the **entire population** to identify trends.*

When to use a Sample:

- When your population is large in size, geographically dispersed, or difficult to contact, it's necessary to use a sample.
- With statistical analysis, you can use sample data to make estimates or test hypotheses about population data.

Example:

- *You're researching **smartphone usage among teenagers** in a city.*
- *The population includes **all teenagers aged 13–18**, which could be tens of thousands.*
- *You select a **random sample of 500 teens** from different schools.*
- *This sample participates in surveys to provide insights into broader usage patterns.*

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Population And Sample Formulas

- Some important formulas related to population and sample are:

Population Parameters:

Mean: The population mean is defined by μ . And its formula is given by,

$$\mu = \frac{1}{N} \sum X, N = \text{Number of elements in population.}$$

Standard Deviation: The population standard deviation is given by σ . And it's formula is given by:

$$\sigma = \sqrt{\frac{1}{N} \sum (X - \mu)^2}$$

Sample Statistic:

Mean: The Sample mean is given by $\bar{x}$. And its formula is given by,

$$\bar{x} = \frac{1}{n} \sum x$$

Standard Deviation: The sample standard deviation is given by s . And it's formula is given by,

$$s = \sqrt{\frac{1}{n-1} \sum (x - \bar{x})^2}$$

Population vs Sample

The main difference between population and sample is given below:

Population	Sample
The population includes all members of a specified group.	A sample is a subset of the population.
Collecting data from an entire population can be time-consuming, expensive, and sometimes impractical or impossible.	Samples offer a more feasible approach to studying populations, allowing researchers to draw conclusions based on smaller, manageable datasets
Includes all residents in the city.	Consists of 1000 households, a subset of the entire population.

Population Parameter vs Sample Statistic

Population Parameter	Sample Statistic
It is a numerical characteristic that describes the entire population	Statistics are calculated from sample data and serve as estimates or approximations of the corresponding population parameters
Parameters are typically unknown and must be estimated.	Calculated using data from a sample drawn from the population. Statistics are directly computed from sample data.
Calculated using data from a sample drawn from the population. Statistics are directly computed from sample data.	Used to estimate population parameters based on sample data. Statistics help researchers infer population characteristics from a representative subset of the population

Example: Estimating Population Height

Suppose you want to determine the **average height of adult males** in a country.

- The **population** includes **all adult males nationwide**.
- The true average height of this population is called the **population parameter** (denoted by μ).

However, measuring the height of every adult male in the country is **impractical**.

To overcome this, you take a **sample**:

- You select **500 adult males** randomly from various regions of the country.
- You measure their heights and calculate the **sample mean height** (denoted by $\bar{x}$).

The **sample mean ($\bar{x}$)** is a **sample statistic**, and it serves as an **estimate of the population mean (μ)**.

Using this sample, researchers can **draw conclusions** about the height distribution of all adult males in th

Importance in CS

- Population and sample are very important in Computer Science especially in fields involving data analysis, machine learning, artificial intelligence, cybersecurity, and more.

1. Data Analysis & Machine Learning

- **Real-world datasets** are often too large (or even infinite), so we work with **samples** to train and test algorithms.
- Example: In training a spam filter, we don't analyze every email ever sent.
- Instead, we use a **sample** of emails to train the model and draw conclusions.

2. Performance Evaluation

- When testing software or systems, we use a **sample of test cases**, users, or inputs to evaluate performance, rather than the entire input space (the population).
- Example: Load testing a server with a **subset of users** to predict performance for the whole user base.

3. Big Data and Cloud Systems

- It's often impossible to process all the data due to **volume and velocity**, so **sampling techniques** are used to summarize trends or build predictive models.

4. Security & Intrusion Detection

- Analyzing **sample network traffic** helps detect suspicious activity without needing to process every packet in real time.
- Helps estimate trends or detect anomalies with fewer computational resources.

Difference Between Qualitative and Quantitative Data

- **Qualitative and Quantitative Data:** Statistics is a subject that deals with the collection, analysis, and representation of collected data.
- The analytical data derived from methods of statistics are used in the fields of geology, psychology, forecasting, etc.



- Quantitative data is numerical, countable, and measurable, providing information on how many, how much, or how often.
- Qualitative data, however, is descriptive, interpretative, and language-based, helping us understand the reasons, processes, or contexts behind certain behaviors.

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What is Qualitative Data?

- The data collected on grounds of categorical variables are qualitative data.
- Qualitative data are more descriptive and conceptual in nature. It measures the data on the basis of the type of data, collection, or category.

- The data collection is based on what type of quality is given.
- Qualitative data is categorized into different groups based on characteristics.
- The data obtained from these kinds of analysis or research is used in theorization, perceptions, and developing hypothetical theories.
- These data are collected from texts, documents, transcripts, audio and video recordings, etc.

Examples of Qualitative Data

- Examples of qualitative data include:
- Textual responses from open-ended survey questions
- Observational notes or fieldwork observations
- Interview transcripts
- Photographs or videos
- Personal narratives or case studies

What is Quantitative Data?

- The data collected on the grounds of the numerical variables are quantitative data.
- Quantitative data are more objective and conclusive in nature. It measures the values and is expressed in numbers.
- The data collection is based on "how much" is the quantity.
- The data in quantitative analysis is expressed in numbers so it can be counted or measured.
- The data is extracted from experiments, surveys, market reports, matrices, etc.

Examples of Quantitative Data

- Some examples of quantitative data are:
- Age, Height, Weight, etc.
- Temperature
- Income
- Number of siblings
- GPA
- Test scores
- Stock

prices

Difference between Qualitative and Quantitative Data

The key differences between Qualitative and Quantitative Data are:

Qualitative vs Quantitative Data	
Qualitative Data	Quantitative Data
Qualitative data uses methods like interviews, participant observation, focus on a grouping to gain collective information.	Quantitative data uses methods as questionnaires, surveys, and structural observations to gain collective information.
Data format used in it is textual. Datasheets are contained of audio or video recordings and notes.	Data format used in it is numerical. Datasheets are obtained in the form of numerical values.
Qualitative data talks about the experience or quality and explains the questions like 'why' and 'how'.	Quantitative data talks about the quantity and explains the questions like 'how much', 'how many' .
The data is analyzed by grouping it into different categories.	The data is analyzed by statistical methods.
Qualitative data are subjective and can be further open for interpretation.	Quantitative data are fixed and universal.

Univariate, Bivariate and Multivariate data and its analysis

- Data analysis is an important process for understanding patterns and making informed decisions based on data.
- Depending on the number of variables involved it can be classified into three main types: univariate, bivariate and multivariate analysis.
- Each method focuses on different aspects of the data which provides a comprehensive understanding of its characteristics and relationships.

Univariate Data

- Univariate data refers to a dataset where each observation is associated with only **one variable**.
- This means it focuses on measuring or observing a single characteristic or attribute for each individual in the dataset.
- It is the most straightforward type of statistical analysis as it finds only one variable at a time without considering relationships with other variables.

Example: Consider the heights (in cm) of seven students in a class.

Heights (in cm)	164	167.3	170	174.2	178	180	186
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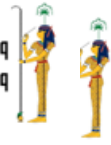
Here the only variable is height and no relationship or interaction with other variables is being considered.

Key points in Univariate analysis

- **1. No Relationships:** It focuses on describing and summarizing the distribution of a single variable.
- It does not explore relationships with other variables or attempt to identify causal connections.

2. Descriptive Statistics

- **2. Descriptive Statistics:** Common techniques include:
- Measures of Central Tendency: Measures such as mean, median and mode to represent the central value of the data.
- Measures of Dispersion: Measures like range, variance and standard deviation to show the spread or variability of the data.



Visualization

- **3. Visualization:** Graphical methods like histograms, box plots and other visual tools are used to represent the distribution and overall pattern of the variable.

Bivariate data

- Bivariate data refers to a dataset where each observation is associated with two different variables.
- The goal of analyzing bivariate data is to understand the relationship or association between these two variables.
- It helps to identify how one variable might affect or be related to the other.

Example: Consider the relationship between temperature and ice cream sales during the summer season:

Temperature	Ice Cream Sales
20	2000
25	2500
35	5000

- In this case, the two variables are temperature and ice cream sales. The data suggests a positive relationship where sales increase as the temperature rises.
- This shows that as one variable (temperature) changes, the other variable (ice cream sales) also changes in a predictable way.

Key Points in Bivariate Analysis

- **Relationship Analysis:** The primary goal of analyzing bivariate data is to understand the relationship between the two variables.
- This relationship could be positive (both variables increase together), negative (one variable increases while the other decreases) or show no clear pattern.

- **Correlation:** Quantifies the strength and direction of the relationship. It ranges from -1 to +1 where +1 shows a perfect positive relationship and -1 shows a perfect negative relationship.
- **Visualization:** Tools like scatter plots help to visualize the relationship between the two variables. Each point on the plot represents a pair of values.

Multivariate data

- Multivariate data refers to datasets where each observation is associated with **more than two** variables.
- This type of data analysis focuses on understanding the interactions and relationships among multiple variables simultaneously.
- It is used when we want to see how multiple variables influence or relate to each other at the same time.

Example:

- Consider a scenario where an advertiser wants to analyze the click rates for different advertisements on a website.
- The data includes multiple variables such as advertisement type, gender and click rate

Advertisement	Gender	Click rate
Ad1	Male	80
Ad3	Female	55
Ad2	Female	123
Ad1	Male	66
Ad3	Male	35

- Here there are three variables: advertisement type, gender and click rate.
- Multivariate analysis allows us to see how these variables interact and how one variable might affect another in the context of the others.

Key Points in Multivariate Analysis

- **Complex Relationships:** Multivariate analysis helps to find patterns and relationships that are not obvious when looking at individual variables.
- **Analysis Techniques:** Techniques such as regression analysis, principal component analysis (PCA) and multivariate analysis of variance (MANOVA) are used to understand interactions between multiple variables.

- **Interpretation:** It provides a deeper and more nuanced understanding of the data which helps us to identify underlying factors that influence the observed patterns.
- There are various tools, techniques and methods available for conducting data analysis including software libraries, visualization tools and statistical testing methods. Each of these can be used to explore and interpret data in various ways.

Real-World Applications

- Each type of data analysis has key applications across various industries:
- **Univariate Analysis**
- **Customer Behavior Studies:** Analyzing a single variable like age or spending patterns helps businesses understand customer demographics and optimize marketing strategies.
- **Financial Reporting:** It is used to track revenue or expenses over time to identify trends.

Bivariate Analysis:

- **Market Research:** Analyzing the relationship between advertising spend and sales helps businesses optimize marketing efforts.
- **Customer Satisfaction:** Exploring the link between product ratings and purchase frequency helps companies understand how satisfaction influences sales.

Multivariate Analysis:

- **Healthcare Predictive Modeling:** Multivariate analysis helps predict patient outcomes based on factors like age, medical history and lifestyle.
- **Financial Risk Analysis:** It's used to assess investment risks by analyzing variables like interest rates, inflation and market conditions.

Difference between Univariate, Bivariate and Multivariate data

Lets see a tabular difference between each of them for better understanding.

Univariate	Bivariate	Multivariate
It summarizes a single variable.	It summarizes two variables.	It summarizes more than two variables.
Does not deal with causes or relationships.	Deals with relationships between two variables.	Analyzes complex relationships between multiple variables.
Does not contain any dependent variable.	Contains one dependent variable.	Contains multiple dependent variables.
The main purpose is to describe the distribution of a single variable.	The main purpose is to explain the relationship between two variables.	The main purpose is to study the relationships among multiple variables.
Example: Height of students in a class.	Example: Temperature and ice cream sales in summer.	Example: Comparison of click rates for advertisements across different genders.

Scales of Measurement

- Data can be classified as being on one of four scales: **nominal, ordinal, interval or ratio**. Each level of measurement has some important properties that are useful to know.
- **Properties of Measurement Scales: Identity** - Each value on the measurement scale has a unique meaning.

- **Magnitude** - Values on the measurement scale have an ordered relationship to one another. That is, some values are larger and some are smaller.
- **Equal intervals** - Scale units along the scale are equal to one another. For Example the difference between 1 and 2 would be equal to the difference between 11 and 12.
- **A minimum value of zero** - The scale has a true zero point, below which no values exist.

1. Nominal Scale

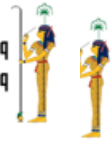
- - Nominal variables can be placed into categories.
- These don't have a numeric value and so cannot be added, subtracted, divided or multiplied.
- These also have no order, and nominal scale of measurement only satisfies the identity property of measurement.
- For example, gender is an example of a variable that is measured on a nominal scale.
- Individuals may be classified as "male" or "female", but neither value represents more or less "gender" than the other.

Ordinal Scale

- - The ordinal scale contains things that you can place in order. It measures a variable in terms of magnitude, or rank.
- Ordinal scales tell us relative order, but give us no information regarding differences between the categories.
- The ordinal scale has the property of both identity and magnitude.
- For example, in a race If Ram takes first and Vidur takes second place, we do not know competition was close by how many seconds.

3. Interval Scale

- - An interval scale has ordered numbers with meaningful divisions, the magnitude between the consecutive intervals are equal.
- Interval scales do not have a true zero i.e In Celsius 0 degrees does not mean the absence of heat. Interval scales have the properties of:
 - Identity
 - Magnitude
 - Equal distance



- For example, temperature on Fahrenheit/Celsius thermometer i.e. 90° are hotter than 45° and the difference between 10° and 30° are the same as the difference between 60° degrees and 80° .

4. Ratio Scale

- - The ratio scale of measurement is similar to the interval scale in that it also represents quantity and has equality of units with one major difference: zero is meaningful (no numbers exist below the zero).
- The true zero allows us to know how many times greater one case is than another.
- Ratio scales have all of the characteristics of the nominal, ordinal and interval scales.

- The simplest example of a ratio scale is the measurement of length.
 - Having zero length or zero money means that there is no length and no money but zero temperature is not an absolute zero.
- Properties of Ratio Scale:
- Identity
 - Magnitude
 - Equal distance
 - Absolute/true zero
 - For example, in distance 10 miles is twice as long as 5 mile

✓ 1. Nominal Data (No order, just categories)

Examples:

Person	Favorite Color	Gender	Type of Pet
A	Blue	Male	Cat
B	Red	Female	Dog
C	Green	Male	Bird
D	Yellow	Female	Fish

➡ Nominal means categories only — no ranking, no numbers with meaning.

✓ 2. Ordinal Data (Categories with order, but spacing unknown)

Examples:

Student	Satisfaction Level	Movie Rating	Education Level
A	Very Satisfied	★★★★☆	High School
B	Satisfied	★★★☆☆	Bachelor
C	Neutral	★★☆☆☆	Master
D	Unsatisfied	★☆☆☆☆	PhD

✓ 3. Interval Data (Numeric, equal intervals, no true zero)

Examples:

Day	Temperature (°C)	IQ Score	Date (Year)
1	20	110	1990
2	25	105	2000
3	30	98	2010
4	15	120	2020

→ Differences matter, but 0 is not meaningful ($0^{\circ}\text{C} \neq \text{no temperature}$).

✓ 4. Ratio Data (Numeric, equal intervals, true zero)

Examples:

Person	Height (cm)	Weight (kg)	Hours Studied
A	170	65	2
B	160	55	4
C	180	72	6
D	165	60	0

Scale	Can be Ordered?	Equal Intervals?	True Zero?	Example
Nominal	✗	✗	✗	Gender, Color
Ordinal	✓	✗	✗	Rating, Rank
Interval	✓	✓	✗	Temperature, Dates
Ratio	✓	✓	✓	Height, Weight, Money

Thank You